BIOPROTEINFORMER: DILATED RESIDUAL TRANSFORMER WITH ARCFACE LOSS FOR FEW-SHOT PROTEIN FAMILY CLASSIFICATION

Software Development Engineer (SDE) — Project Showcase

Candidate Details

Field	Details
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GitHub Repository	https://github.com/sivaanjan/BioProteinFormer
Portfolio	https://portfolio-kappa-five-a3r49h40bt.vercel.app/
Project Type	Deep Learning, Bioinformatics, Few-Shot Learning

1. Project Overview

Every protein in a cell has a job, and the fastest way to guess that job is to ask which family the protein belongs to. Over 250 million protein sequences are known, but fewer than 600,000 have experimentally verified functions. For rare enzyme subfamilies or newly discovered viral proteins, a scientist may have fewer than 20 confirmed examples to learn from.

Protein language models (ESM-2, ProtTrans) are powerful but heavy to run and still hungry for labelled data. This project asks a sharper question: **can a small, purpose-built network recognise a protein family after seeing just 1 to 10 examples?**

The answer is **BioProteinFormer**, a compact network trained from scratch. It reads each amino acid through three lenses (identity, biochemistry, local context), scans the sequence with a dilated convolutional tower, adds attention for global context, and is trained with an ArcFace angular-margin loss so that every family forms a tight, well-separated cluster on a hypersphere. A new protein is classified by finding the family prototype it is closest to.

The system was built from live data collection through model training, benchmarking against ten baselines, ablation, and embedding visualisation, all under one identical and reproducible protocol. It reached **89.40%** test accuracy on 12 families, **55.2%** 5-way accuracy from a single example per class, and **79.9%** from ten examples, leading all ten baselines by **2.2 to 11.1 percentage points**.

2. Problem Statement and Motivation

Proteins are not images, and few-shot methods built for images do not transfer cleanly. Table 1 lists the five practical difficulties, each of which motivated a concrete design decision:

Problem	Why it is hard	Solution Implemented
Variable length	Sequences range from dozens to thousands of residues	Fixed length 512 with truncation and padding
Long-range dependencies	Residues far apart in sequence interact in 3-D	Multi-head self-attention over the whole sequence
Multi-scale structure	Motifs (3–20), domains (50–300), global fold	Dilated convolutions with rates 1, 2, 4, 8, 16
Biochemical grounding	Amino acids carry meaning beyond their letter	7 physicochemical channels + bigram log-odds
Sparse class signatures	A family may hinge on one short motif	SE channel attention + max pooling to catch motifs anywhere

The project was framed not as a single model but as a complete, reproducible machine-learning system: data acquisition, leakage-free preprocessing, model design, fair benchmarking, few-shot evaluation, and ablation.

3. Project Objectives

- Build a compact encoder for few-shot protein family classification, trained from scratch with no pretrained language model.
- Give the model domain knowledge through multi-modal per-residue inputs instead of relying on millions of training sequences.
- Shape the embedding space with ArcFace so nearest-prototype matching works even with a single support example.
- Compare against ten baselines under one identical training and evaluation protocol.
- Run ablations to show which components genuinely matter, and visualise the embedding space with t-SNE.

4. Technologies and Tools Used

Layer	Technology	Version / Notes
Deep learning	TensorFlow / Keras	2.16.1 (GPU), Python 3.10.14
Compute	Kaggle cloud	NVIDIA Tesla T4 (16 GB), single GPU
Data access	RCSB PDB Search and Data REST APIs	Requests 2.31, tqdm
Data science stack	NumPy, Pandas, scikit-learn	NumPy 1.26.4, Pandas 2.2.1, scikit-learn 1.4.1 (metrics, t-SNE)
Visualisation	Matplotlib, Seaborn	3.8.4 / 0.13.2: EDA, curves, confusion matrices
Optimiser	AdamW, cosine warm restarts	lr 5e-4, weight decay 1e-4

5. System Architecture and Model Pipeline Design

The model follows a strict *Read → Scan → Attend → Embed → Match* paradigm. Each residue is fused from three input lenses into 40 dimensions, lifted to 128 channels, scanned at multiple scales, re-weighted by channel and self-attention, pooled to 256 dimensions, and placed on the unit hypersphere. ArcFace shapes that space during training; cosine prototype matching uses it at inference.

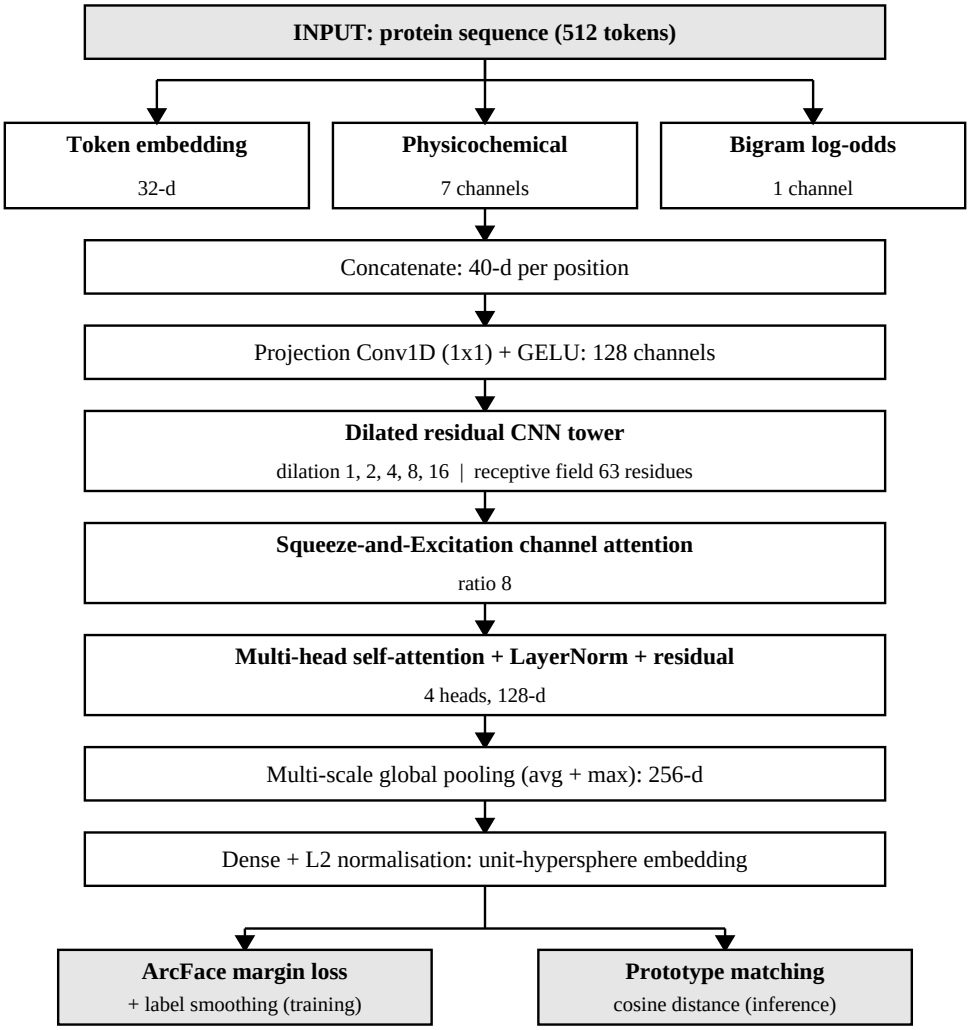


Figure 1: End-to-end model pipeline — three-lens input encoding, dilated residual tower, attention, unit-sphere embedding, ArcFace training and prototype matching.

The layered design yields three system-level properties: compactness (no pretrained model, single T4 GPU), multi-scale coverage (local motifs to global context in one encoder), and a training objective matched to how the model is used at inference.

6. Dataset Description

Instead of a ready-made benchmark, the dataset was built live from the RCSB Protein Data Bank. For 12 functional families the PDB was queried through its REST API (filter: polymer type = Protein, keyword = family name). Canonical sequences were kept only if they contained the 20 standard amino acids and had length 40–1024, giving about 500 sequences per family.

The 12 families: Hydrolase, Transferase, Oxidoreductase, Immune System, Lyase, Transcription, Signaling Protein, Transport Protein, Viral Protein, Isomerase, Ligase, Membrane Protein.

Split	Sequences	Purpose
Train	4,080 (68%)	Learn the embedding
Validation	720 (12%)	Early stopping
Test	1,200 (20%)	Final scores and few-shot episodes

Exploratory findings. Membrane proteins were enriched in Leu, Val, Ile, Ala and Phe (hydrophobic residues spanning a lipid bilayer). Immune proteins were rich in Cys and Tyr (disulfide bonds, antigen-binding sites). Hydrolases showed Ser and His (the catalytic triad). These signatures are exactly what the physicochemical input

channels are designed to capture. A k-mer analysis showed coverage collapsing fast (about 80% of 2-mers, 25% of 3-mers, 6% of 4-mers), which is why the model reads single amino acids rather than n-gram tokens.

7. End-to-End Implementation

7.1 Data Acquisition from RCSB PDB

A Python collector queries the RCSB Search API per family keyword, retrieves canonical sequences through the Data API, filters by alphabet and length, and de-duplicates per family. The result is a balanced 6,000-sequence dataset split 68/12/20 with a fixed seed (42) for reproducibility.

7.2 Per-Residue Input Encoding (40 dimensions)

- **Token embedding (32-d):** a trainable vector for amino-acid identity (vocabulary of 22: 20 amino acids + PAD + UNK).
- **Physicochemical features (7-d):** Kyte–Doolittle hydrophobicity, net charge, aromaticity, molecular weight, isoelectric point, Chou–Fasman beta-sheet propensity, van der Waals volume, z-score normalised on training data only.
- **Bigram log-odds (1-d):** how much more often the previous and current residue occur together than chance, capturing motifs such as Gly-Pro and Cys-Cys.

7.3 Encoder Architecture

Table 2 summarizes the stages of the encoder, applied in order to every protein sequence.

Stage	Layer	Role
Projection	1x1 Conv + GELU, 40 → 128	Brings every later stage to one width
Local-to-regional scan	5 residual blocks, 3-wide conv, dilation 1, 2, 4, 8, 16 + BatchNorm + GELU	Receptive field = 1 + (3−1)(1+2+4+8+16) = 63 residues; covers helices (7–15), beta strands (5–12), binding loops (20–40), interfaces (30–50)
Channel attention	Squeeze-and-Excitation, MLP 128 → 16 → 128, sigmoid gates	Amplifies channels carrying family-specific motifs, damps noisy ones
Global reader	Multi-head self-attention (4 heads, 128-d) + LayerNorm + residual	Any residue attends to any other, e.g. catalytic residue to cofactor site 300 positions away
Readout	Global average + global max pooling (256-d), Dense, L2 norm	Composition + strongest motif; embeddings placed on the unit hypersphere

7.4 ArcFace Loss

Standard cross-entropy builds good classifiers for seen classes but not necessarily a well-organised space for new ones. ArcFace adds an angular margin (m = 0.5 rad, about 28.6°, scale s = 64) to the true class before the softmax, so each family must be separated from the others by a guaranteed angle. Combined with label smoothing (0.1), this yields compact clusters.

7.5 Prototypical Few-Shot Inference

At test time the mean embedding of K support sequences becomes a class prototype, and a query takes the label of the nearest prototype by cosine distance:

```
P(y | x_q, S) = softmax( -d( phi(x_q), c_k ) ), c_k = mean of phi(x) over the K support sequences of class k
```

7.6 Training Configuration

Item	Setting
Sequence length / vocabulary	512 / 22 tokens (20 amino acids + PAD + UNK)
Optimiser and schedule	AdamW, lr 5e-4, weight decay 1e-4, cosine warm restarts (T0 = 5, Tmult = 2)
Batch size / epochs	64 / up to 30, early stopping patience 5 on validation accuracy
Regularisation	Dropout 0.35 (classifier head), label smoothing 0.1
Seed	42 (reproducible)
Few-shot protocol	5-way, K in {1, 2, 3, 5, 10}, 5 queries per class, 100 episodes, mean ± 95% CI

8. Key Engineering Decisions and Design Considerations

- **Compact and from scratch over a huge pretrained language model:** easier to train, deploy and reason about.
- **Dilated residual tower over plain deep CNNs:** exponential receptive field with linear parameter growth.
- **CNN first, one attention block after:** global context at a fraction of full-transformer cost.
- **ArcFace over cross-entropy:** the loss is chosen to match how the model will be used (nearest-prototype matching).
- **Character-level tokens over k-mers:** backed by the k-mer coverage analysis.
- **Train-only normalisation statistics:** feature means and deviations are computed on training sequences only and applied to validation and test, avoiding leakage.
- **Identical baseline protocol:** all ten baselines share the same 32-d embedding, 128-d output, optimiser, early stopping and label smoothing, so only the sequence-processing architecture (and ArcFace for BioProteinFormer) differs.

9. Experimental Setup and Evaluation Methodology

All experiments ran on a single NVIDIA Tesla T4 on Kaggle with TensorFlow/Keras 2.16.1. Supervised performance was measured on the held-out test set of 1,200 sequences using accuracy, macro F1, precision and recall. Few-shot performance was measured with 5-way K-shot episodes drawn from the test set (K in {1, 2, 3, 5, 10}, 5 queries per class, 100 episodes) and reported as mean ± 95% confidence interval; random chance is 20%. Ten baselines (Simple CNN, Deep CNN, TextCNN, Dilated CNN, BiLSTM, GRU+Attention, CNN-BiLSTM, ResNet-1D, SE-CNN, DPCNN) were trained under the same protocol.

10. Results and Performance Analysis

10.1 Supervised Classification

The full run on a single Tesla T4 produced the headline results in Table 3. BioProteinFormer ranked first on every supervised metric on the 1,200-sequence test set, with a lead of 2.19 points over the runner-up (ResNet-1D) and 11.06 points over the weakest baseline.

Metric	Value
Test accuracy (12 families)	89.40%
Macro F1 / Precision / Recall	89.38% / 89.52% / 89.40%
5-way accuracy, K=1 (chance 20%)	55.2% (2.76x chance)
5-way accuracy, K=10	79.9%
Lead over ten baselines	+2.2 to +11.1 pp
Strongest family	Membrane Protein, recall above 97%

Rank	Model	Acc (%)	F1 (%)	Prec (%)	Recall (%)
1	BioProteinFormer	89.40	89.38	89.52	89.40
2	ResNet-1D	87.21	87.19	87.33	87.21
3	SE-CNN	86.85	86.89	87.02	86.85
4	CNN-BiLSTM	86.44	86.41	86.55	86.44
5	Deep CNN	85.92	85.88	86.01	85.92
6	Dilated CNN	85.76	85.73	85.88	85.76
7	GRU+Attention	85.21	85.18	85.31	85.21
8	BiLSTM	84.93	84.90	85.04	84.93
9	TextCNN	83.67	83.61	83.79	83.67
10	DPCNN	82.88	82.85	83.01	82.88
11	Simple CNN	78.34	78.28	78.47	78.34

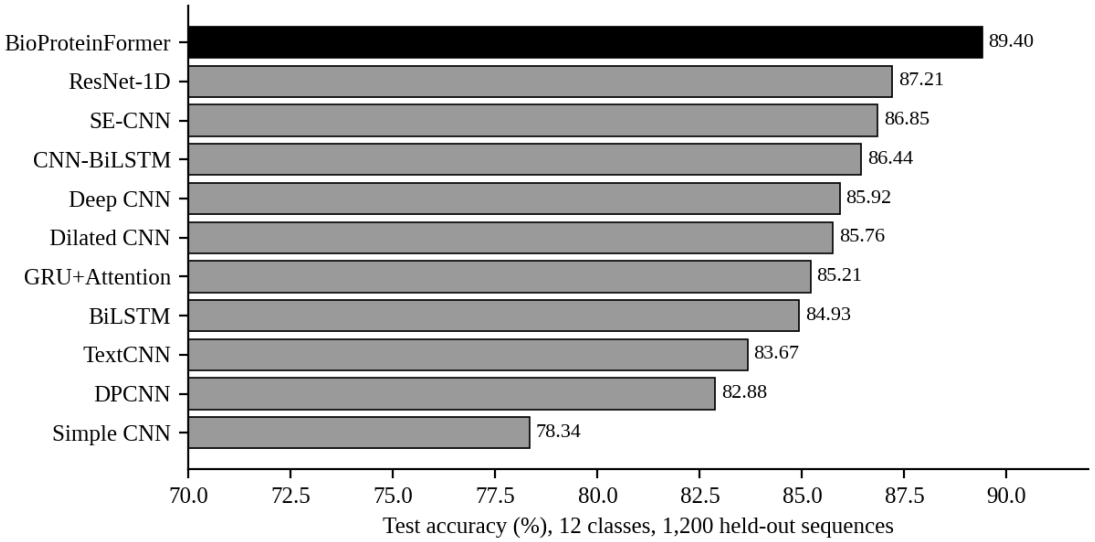


Figure 2: Supervised test accuracy — BioProteinFormer versus ten baselines on 1,200 held-out sequences across 12 families.

Models with skip connections or channel attention beat plain sequential CNNs and recurrent models, suggesting that how features are aggregated matters more than recurrence for protein embeddings.

10.2 Few-Shot Classification

Table 5 gives 5-way K-shot accuracy. With one example per family, BioProteinFormer identifies the right family for more than half of queries, 2.76 times better than chance.

Model	K=1	K=3	K=5	K=10
BioProteinFormer	55.2 ± 3.2	68.8 ± 2.9	74.6 ± 2.7	79.9 ± 2.4
ResNet-1D	52.4 ± 3.4	65.1 ± 3.1	70.9 ± 2.9	76.2 ± 2.7
SE-CNN	51.8 ± 3.5	64.4 ± 3.2	70.1 ± 3.0	75.6 ± 2.8
CNN-BiLSTM	50.9 ± 3.6	63.5 ± 3.3	69.0 ± 3.1	74.4 ± 2.9
Deep CNN	47.1 ± 3.9	59.4 ± 3.6	64.6 ± 3.4	69.8 ± 3.2
Simple CNN	38.2 ± 4.3	48.8 ± 4.1	53.4 ± 4.0	58.6 ± 3.8

5-way K-shot accuracy (%), mean ± 95% CI over 100 episodes; chance = 20%. Selected models shown.

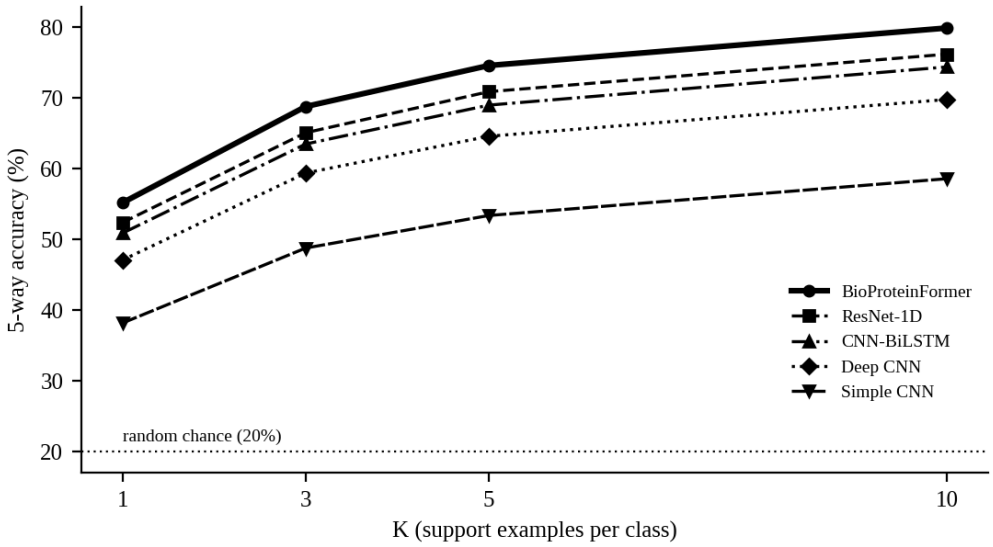


Figure 3: Few-shot accuracy versus number of support examples — 5-way episodes, 100 per setting, chance = 20%.

- **Leads at every K.** The advantage is architectural, not a lucky shot count.
- **Biggest jump from K=1 to K=3** (+13.6 pp), as prototype variance drops fastest there.
- **Embedding quality beats quantity.** Simple CNN with 10 shots (58.6%) is below BioProteinFormer with just 3 shots (68.8%).

10.3 Embedding Space and Error Analysis

t-SNE on 1,000 test embeddings shows BioProteinFormer forming compact family clusters, while weaker baselines produce diffuse, overlapping clouds, mirroring the accuracy ranking. The remaining mistakes are biologically sensible, a good sign that the model learned real structure:

- Hydrolase confused with Ligase (shared nucleotide-binding motifs).
- Transcription confused with Signaling Protein (shared helix-turn-helix domains).
- Oxidoreductase confused with Transferase (shared Rossmann fold).
- Membrane Protein is almost perfect (over 97% recall), driven by its hydrophobic composition captured by the physicochemical channels.

10.4 Ablation Study

Table 6 reports the ablation: each component was removed or swapped in isolation while everything else stayed fixed.

Configuration	Acc (%)	K=5 FSL (%)	Δ FSL
Full BioProteinFormer	89.40	74.56	—
w/o physicochemical channels	86.12	70.24	−4.32
w/o bigram channel	88.97	74.01	−0.55
w/o dilated tower (1 Conv1D)	84.78	67.83	−6.73
w/o SE channel attention	87.23	72.14	−2.42
w/o multi-head self-attention	87.84	72.89	−1.67
ArcFace → CrossEntropy	86.91	68.32	−6.24
ArcFace → CosFace	88.73	73.82	−0.74
w/o label smoothing	88.91	73.94	−0.62
GAP-only pooling	88.34	73.11	−1.45

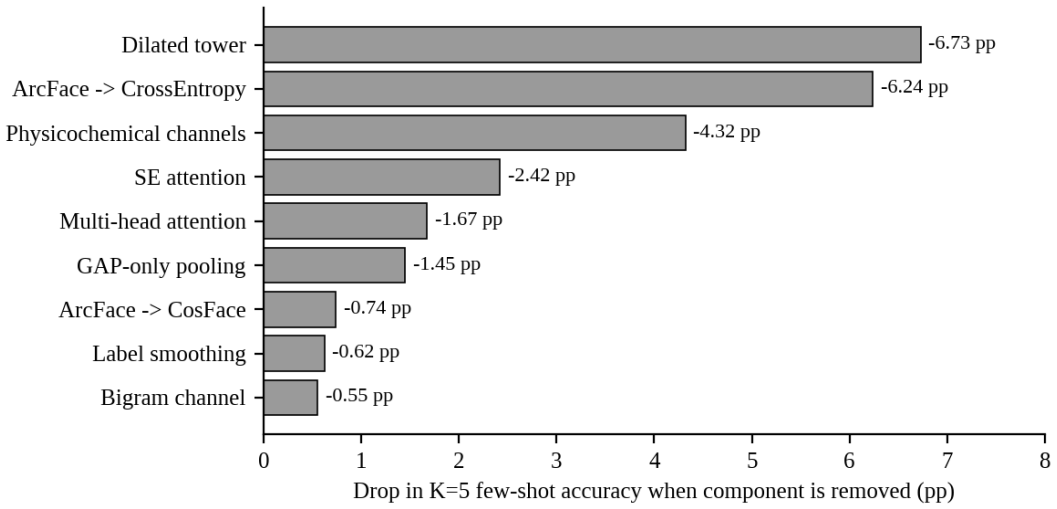


Figure 4: Ablation study — loss in K=5 few-shot accuracy when each component is removed (full model = 74.56%).

- **Dilated tower (−6.73 pp):** the multi-scale receptive field is the backbone of the model.
- **ArcFace (−6.24 pp):** swapping to cross-entropy costs only 2.49 pp of supervised accuracy but 6.24 pp of few-shot accuracy, showing the loss improves embedding geometry, not just classification. CosFace performs almost as well (−0.74 pp), so the angular-margin idea is what counts.
- **Physicochemical channels (−4.32 pp):** domain knowledge adds more than token identity alone.

11. Challenges Faced and Solutions Implemented

Challenge	Solution
Few labelled examples per family	Compact encoder with biochemical inputs; ArcFace-shaped embeddings and nearest-prototype matching
Sequences of widely varying length	Fixed length 512 with truncation and padding
Local motifs and global interactions both matter	Dilated residual tower (RF = 63) followed by multi-head self-attention
Sparse k-mer coverage (80% / 25% / 6% for 2/3/4-mers)	Character-level (single amino acid) tokens
Feature-statistics leakage	Normalisation means and deviations computed on training sequences only
Unfair model comparison	Identical embedding size, output width, optimiser, early stopping and label smoothing across all baselines
Cross-entropy gave poorly organised few-shot space	ArcFace angular margin (m = 0.5, s = 64) plus label smoothing

12. My Technical Contributions

This was an individual project. I designed and implemented:

- The live PDB data-collection pipeline (RCSB REST APIs), filtering, balancing and leakage-free train/validation/test splitting.
- The multi-modal per-residue input encoding: token embedding, seven physicochemical channels and bigram log-odds.
- The BioProteinFormer encoder: dilated residual tower, SE channel attention, multi-head self-attention and multi-scale pooling, with ArcFace training and prototypical few-shot inference.
- Ten baseline models under one identical protocol, with metrics, few-shot episode evaluation (100 episodes, 95% CI), t-SNE visualisation and confusion analysis.
- The full ablation study isolating each component’s contribution.

13. Limitations and Future Improvements

- **Benchmark scope:** 12 broad families; fine-grained subfamilies and truly uncharacterised proteins remain untested.
- **Sequence only:** no 3-D structure information (secondary structure, AlphaFold2 contact maps could help).
- **No language-model comparison:** ESM-2 and ProtBert were not benchmarked due to compute.
- **No episodic meta-training:** the model is trained supervised and only evaluated episodically; MAML or Prototypical-Network training is a natural next step.
- **Future directions:** ESM-2 features as an extra input channel, structure-aware inputs, transductive prototype refinement, multi-label families, and large benchmarks such as Pfam, SCOP and CATH.

14. Project Repository and Documentation

All code, notebooks and documentation are public at https://github.com/sivaanjan/BioProteinFormer (MIT licence). The repository contains:

- **notebooks/** — the Jupyter notebooks for data collection, training, baselines, few-shot evaluation and ablation; the README names **Final_code.ipynb** as the notebook to open and run end to end.
- **Reports/** — the written project report and supporting documents.

- **README.md** — overview, dataset statistics, architecture summary, headline results, installation and running instructions.
- **Requirement.txt** — Python dependencies, installed with pip.
- **LICENSE** and **.gitignore** — MIT licence and repository hygiene.

To reproduce the results, clone the repository, install the requirements, open Final_code.ipynb and run all cells; the seed is fixed at 42.

15. Conclusion

BioProteinFormer shows that careful design can beat sheer size in the data-scarce regime. Three ideas work together: give the model biochemical knowledge (multi-modal inputs), give it the right field of view (dilated tower plus attention), and shape the space it learns (ArcFace on a hypersphere). The result is 89.40% supervised accuracy, 55.2% from a single example per class and 79.9% from ten, ahead of ten baselines at every shot count, with ablations that pinpoint why.

Beyond the metrics, the project exercised core software-engineering skills: building a data pipeline from a live API, preventing leakage, designing a model around the way it will be used, benchmarking fairly, and documenting results so another engineer can reproduce them. For rare protein families where labels are scarce, it is a practical step towards faster functional annotation.

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